Vilém Zouhar

vilem.zouhar@gmail.com · vilda.net

Education

ETH Zürich 2022-Dec 2026 (expected)
PhD in Computer Science in **LRE lab**
Advisors: Mrinmaya Sachan, Menna El-Assady
Thesis: Prudent & Bespoke AI Evaluation

Saarland University + Groningen University Sep 2020 - Jun 2022
MSc. in Language Science and Technology (double)
Advisors: Dietrich Klakow, Gosse Bouma
Thesis **Shrinking Knowledge Base Size: Dimension Reduction & Filtering**

Charles University in Prague Sep 2017 - Jun 2020
BSc. in Computer Science (computational linguistics minor)
Advisor: Ondřej Bojar
Thesis **Enabling Outbound Machine Translation**

Research Interests (inter alia)

- Formal methods of AI evaluation and safety — *„Without evaluation, science is limited to blind exploration.“*
- Practical AI evaluation also with humans — *„How to make humans judge quality?“*
- Efficient AI evaluation with applied statistics — *How to compare models robustly yet economically?*

Work Experience

- **Google Translate**, Mountain View (research intern, evaluation) Jun-Oct 2025
- Microsoft Translate, remote (applied scientist intern, model evaluation) Jun-Oct 2024
- **Amazon Machine Translation**, New York (applied scientist intern, quality estimation) Jun-Oct 2023
- **Spoken Language Systems**, Saarland University (student research assistant) Jan-Dec 2021
- **Institute of Formal and Applied Linguistics**, Charles University (student research assist.) Oct 2019 - Jun 2022
- Previo (software dev intern) Jun-Sep 2018
- BIM Project (software dev intern) Jun-Sep 2017

Awards and Scholarships

- Cohere Labs Catalyst grant for efficient evaluation 2026
- Google PhD Fellowship in Natural Language Processing 2025
- Two EAMT grants for research on compute-efficient evaluation, and LLM brittleness 2025, 2025
- Language & Communication Technologies full MSc Erasmus Mundus **scholarship** 2020-2022
- Student faculty grant for **learning materials** for Machine Learning Exercises in R 2018

Teaching

- Human evaluation tutorial EAMT 2026, KONVENS 2026, MT Marathon 2026
- Large Language Models TA **2023**, **2024**, **2025**, **2026** (+lecture)
- Deep Learning TA **2025**
- Interactive Machineimon Su Learning Visualization and Explainability TA **2025**
- Neural Networks Implementation and Applications head TA **2021**
- Statistical Natural Language Processing head TA **2021**

Invited talks

- Theory and Practice of of Efficient Evaluation at Google Gboard (2026), SICSA (2026)
- Selecting Examples to Human-Evaluate NLG Models at Charles University (2025), TU Darmstadt (2025), UMD Multilingual Seminar (2026)
- Prudent NLG Evaluation at KIT (2025), Cardiff NLP Seminars (**2025**), Google Translate (2024)
- Token(s) of Appreciation for BPE at MT Marathon (**2024**)
- ESA and ESAAI at Microsoft Translate (2024)
- How we solved tokenization but got it wrong at ZurichNLP Meetup #9 (**2024**)
- Quality and Quantity of Machine Translation References for Automated Metrics at IST/Unbabel seminar (**2024**)
- Poor Man's Quality Estimation at IST/Unbabel seminar (**2023**)

Service

- **General Machine Translation** Shared Task at WMT (co-organizer) 2024, 2025, 2026
- **Evaluation Metrics** Shared Task at WMT (co-organizer) 2025, 2026
- **Multilingual Instruction** Shared Task at WMT (co-organizer) 2025, 2026
- **Multilingual Multicultural Evaluation Workshop** at EACL (organizer) 2026
- **SIGTYP Workshop: Computational Linguistic Typology and Multilingual NLP** at EACL (co-organizer) 2026
- **ACL Mentorship Panel**: Navigating Mental Health in Fast-Paced AI Research (panelist) 2026
- **WMT Terminology** Shared Task (organizer) 2023, 2025
- Evaluation committee member for granting university accreditations in Czechia 2020-2025
- **CSRR**: Workshop on Commonsense Representation and Reasoning (organizer) ACL 2022
- Area chair GEM 2026, EMNLP Demo 2026, ARR 2026
- Reviewer: {EACL, MME, ACL, ARR, SwissText, GEM, EMNLP Demo, ARR}2026 {EAMT, CHI} 2027 {NAACL★, ACL, EMNLP, WMT, ARR}2025 {LREC-COLING, EACL★, ARR, Repl4NLP, EMNLP, WMT}2024 {ARR, WMT, EMNLP, AACL-IJCNLP, ACL}2023 {SVRHM, CoNLL, AACL-IJCNLP}2022

Technical Knowledge

GitHub

- **Programming:** Python, JS/TS, Rust
- **Toolkits:** LM eval harness, vllm, PyTorch, Scikit, Numpy, HF, Fairseq, Marian
- **Misc:** Linux (long-term user, cluster), passionate about visualization, writing and typesetting (Typst, LaTeX)

Languages

- Czech: C2
- English: C2
- German: B2

+ linguistic curiosity

Students

Pleasure to (co-)supervise student projects/theses

- Rosamund Ang: Sample-Efficient Aligning of Automatic Metrics ETH 2026
- Šimon Sukup, William Kalikman, Michal Tešnar: Augmenting Texts to Increase Translation Difficulty ETH 2026
- Victor Zarzu: Speech Translation Quality Estimation ETH 2026
- Chenfei Xiong: Collaborative Discovery of Edge Cases in Text Classification ETH 2025
- Oymak Gül, Hatipoglu Öykü, Garries Urbina Oscar, Victor Zarzu Interactive Visualization of Chain-of-Thought Reasoning in Large Language Models ETH 2025
- Yijie Tong, Haokun He (Self Quality Estimation for Machine Translation and NLP) UZH 2024
- Abhinav Kumar (Data Augmentation for Text Complexity Prediction) UZH 2024
- Vincent Dörig (IRead: AI-Enhanced Textbook Reading) ETH 2024
- David Gu (Manifestations of Image Complexity) ETH 2023

Extra-Curricular

- Academic senate at Charles University Faculty of Mathematics and Physics 2018-2020
- Several **games** programmed and presented in limited time, mostly Ludum Dare 2015-present
- Organization of **Kasiopea**, an annual coding competition for talented high school students 2017-2019
- Amateur electric guitar 2021-present